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Graphic Design with Large Multimodal Model

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Abstract. In the field of graphic design, automating the integration of design elements into a cohesive multi-layered artwork not only boosts productivity but also paves the way for the democratization of graphic design. One existing practice is Graphic Layout Generation (GLG), which aims to layout sequential design elements. It has been constrained by the necessity for a predefined correct sequence of layers, thus limiting creative potential and increasing user workload. In this paper, we present Hierarchical Layout Generation (HLG) as a more flexible and pragmatic setup, which creates graphic composition from ***unordered*** sets of design elements. To tackle the HLG task, we introduce Graphist, the first layout generation model based on large multimodal models. Graphist efficiently reframes the HLG as a sequence generation problem, utilizing RGB-A images as input, outputs a JSON draft protocol, indicating the coordinates, size, and order of each element. We develop multiple evaluation metrics for HLG. Graphist outperforms prior arts and establishes a strong baseline for this field. Project homepage: https://github.com/graphic-design-ai/graphist

Keywords: Graphic design · Layout generation · LMM · MLLM

1 Introduction

Graphic design [41] fundamentally serves as a form of visual communication. It involves the creation and combination of symbols, images, and text to express certain ideas or messages. This field requires significant expertise and time investment to produce aesthetically pleasing graphic compositions. Recently, there is a significant paradigm shift in leveraging AI for automating the layout of given design elements into cohesive graphic compositions heralds [26, 27, 90, 100]. This could potentially reduce the workload of professional designers and provide an avenue for beginners to create their design pieces, making graphic design more democratize and effective.

A preliminary attempt to automate this process is observed in the task known as **G**raphic **L**ayout **G**eneration (GLG) [16,25,30,35,39,40,52–54,83,86,97]. GLG

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Fig. 1: Schematic diagram of hierarchical layout generation. (Left) A comparison between the traditional *GLG* task and the newly proposed *HLG* task, with the major difference in that HLG relaxes the constraint of GLG, so that unordered multimodal input elements can be processed. (Right) Errors in either layer sequencing or spatial positioning can significantly impact the overall quality of the design.

attempts to intelligently arrange provided elements into attractive compositions under the assumption of a *predefined order of layers*. However, establishing the appropriate ordering of these layers is a design cornerstone that, if mismanaged, can fracture the visual hierarchy, leading to disarray in the intended message delivery. Requiring users to prescribe an accurate layer sequence prior to layout not only burdens them with foresight and planning but also stifles layout algorithms, restricting their capacity to transcend such confines in the pursuit of innovative and aesthetically superior outcomes.

In pursuit of more practical applications, this paper introduces a new task, **H**ierarchical **L**ayout **G**eneration (HLG). HLG to craft a visually appealing graphic composition from a collection of unordered elements by meticulously considering both their spatial arrangement and the sequencing of layers. The adjective ***hierarchical*** emphasizes the significance of element ordering, setting HLG apart from conventional layout generation practices. The distinction between GLG and HLG is illustrated in the left panel of Figure 1, while the right panel of the same figure captures the impact of layer sequencing and spatial positioning on the visual aesthetics of graphic design.

For the HLG task, we introduce Graphist, the first layout generation model built upon Large Multimodal Model (LMM) [13, 18, 49, 57, 70, 104]. The layout generation task is challenging that it digests diverse input elements such as RGB-A materials, RGB images, and texts, and the desired outcomes must precisely reflect the intricate relationships among these multimodal input elements. LMMs are well-suited for this task, as they can unify different modalities, like images, text, coordinates [13, 64] into tokens. This allows for flexible configuration of various tasks, such as HLG, GLG, and more variants. The knowledge and strong reasoning ability stored in pre-trained model parameters are also beneficial for layout tasks. Furthermore, LMMs demonstrate significant potential for scaling [42, 61, 73], enabling the pursuit of enhanced performance through the use of larger models and more extensive datasets. For these reasons, LMMs were

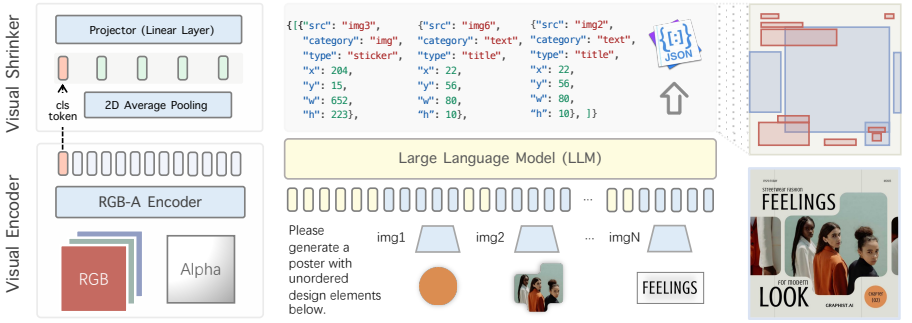

Fig. 2: Graphist Pipeline. Graphist comprises three components: RGB-A Encoder, Visual Shrinker, and a LLM. It accepts a variety of design elements and generates a graphic composition in JSON format end-to-end.

a natural choice for our foundational architecture in developing Graphist. In our specific approach for HLG, we train Graphist with graphic composition data, in which, as shown in Figure 2, each design element is represented as an RGB-A image as input, the model then generates a JSON draft protocol, which specifies the coordinates, size, order and other attributes of each design elements in end-to-end manner.

To evaluate the HLG task, we introduce two novel metrics: Inverse Order Pair Ratio (IOPR) and GPT-4V Eval. The former assesses the accuracy of the layer order in the graphic composition, while the GPT-4V Eval, leveraging the capabilities of GPT-4V [62], quantifies the overall aesthetic quality. Additionally, we have incorporated a human rating score to align with the subjective perceptions of real individuals. Graphist has emerged as the SoTA solution, excelling across all these metrics. Furthermore, Graphist consistently delivers impressive performance on conventional GLG task.

We summarize our contributions as follows:

- We introduce the Hierarchical Layout Generation (HLG) task, which creates graphic compositions from unordered design elements. HLG overcomes the constraints of the Graphic Layout Generation that requires pre-determined layer ordering. The new setup allows more flexible and practical AI-assisted graphic designs.
- We present Graphist, the first LMM-parameterized layout generation model that can be trained end-to-end. Graphist accepts a variety of vision-and-language multimodal design elements and generates a graphic compositions in JSON code format, which can be further automatically rendered into the final design.
- We develop evaluation metrics for HLG, including IOPR and GPT-4V Eval. Graphist shows superior performance in these metrics, setting a robust benchmark for the field.